№12

:

2026-03-19

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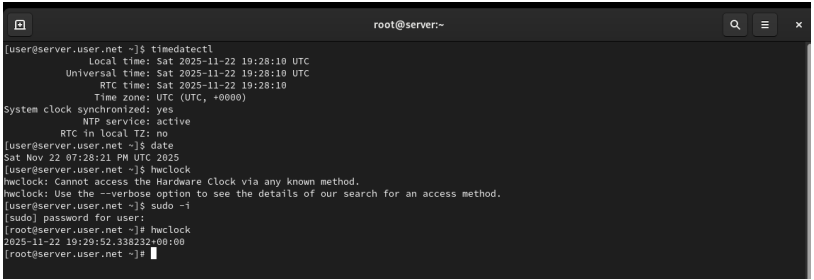
2.

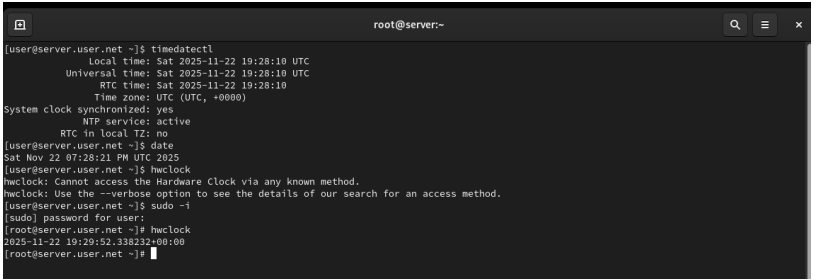
- `timedatectl`



1: timedatectl

2.1

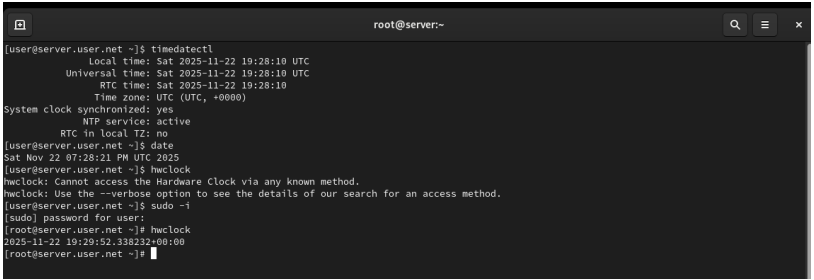
- `timedatectl`
- UTC (+0000)



1: timedatectl

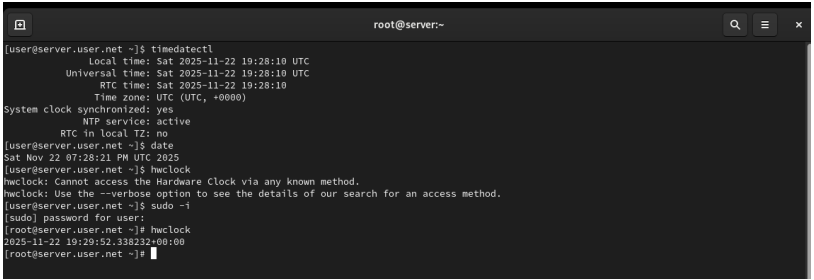
2.1

- `timedatectl`
- UTC (+0000)
-



1: timedatectl

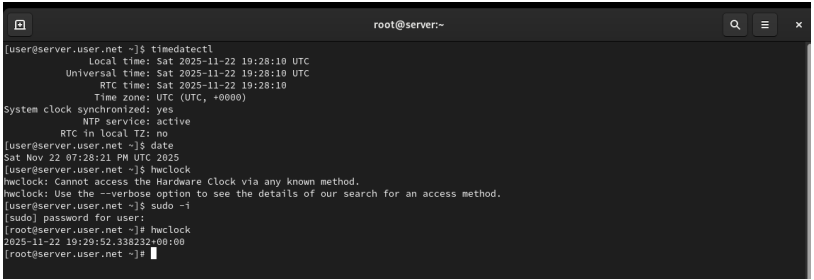
- `timedatectl`
- UTC (+0000)
-
-



1: timedatectl

2.1

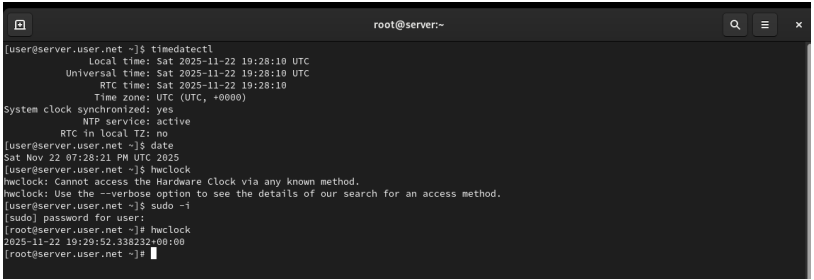
- `timedatectl`
- UTC (+0000)
-
-
- NTP



1: timedatectl

2.1

- timedatectl
- UTC (+0000)
-
-
- NTP
- RTC UTC



1: timedatectl

- `date`

- date
- UTC

- date
- UTC
- hwclock

- date
- UTC
- hwclock
- RTC

- `date`
- UTC
- `hwclock`
- RTC
- UTC (+00:00)

2.2

- chrony

```
[root@server.user.net ~]# systemctl restart chronyd
[root@server.user.net ~]# firewall -cmd --add-service=ntp --permanent
bash: firewall: command not found...
[root@server.user.net ~]# firewall -cmd --add-service=ntp  --permanent
bash: firewall: command not found...
[root@server.user.net ~]# irewall-cmd --add-service=ntp --permanent
bash: irewall-cmd: command not found...
[root@server.user.net ~]# firewall-cmd --add-service=ntp --permanent
success
[root@server.user.net ~]# firewall-cmd --reload
success
[root@server.user.net ~]#
```

2: chrony dnf

2.2

- chrony
-

```
[root@server.user.net ~]# systemctl restart chronyd
[root@server.user.net ~]# firewall -cmd --add-service=ntp --permanent
bash: firewall: command not found...
[root@server.user.net ~]# firewall -cmd --add-service=ntp  --permanent
bash: firewall: command not found...
[root@server.user.net ~]# irewall-cmd --add-service=ntp --permanent
bash: irewall-cmd: command not found...
[root@server.user.net ~]# firewall-cmd --add-service=ntp --permanent
success
[root@server.user.net ~]# firewall-cmd --reload
success
[root@server.user.net ~]#
```

2: chrony dnf

2.2

- chrony
-
-

```
[root@server.user.net ~]# systemctl restart chronyd
[root@server.user.net ~]# firewall -cmd --add-service=ntp --permanent
bash: firewall: command not found...
[root@server.user.net ~]# firewall -cmd --add-service=ntp  --permanent
bash: firewall: command not found...
[root@server.user.net ~]# irewall-cmd --add-service=ntp --permanent
bash: irewall-cmd: command not found...
[root@server.user.net ~]# firewall-cmd --add-service=ntp --permanent
success
[root@server.user.net ~]# firewall-cmd --reload
success
[root@server.user.net ~]#
```

2: chrony dnf

- chronyc sources

```
[root@client.user.net ~]# chronyc sources
MS Name/IP address         Stratum Poll Reach LastRx Last sample
===============================================================================
[root@client.user.net ~]#
```

3: (chronyc sources)

- chronyc sources
-

```
[root@client.user.net ~]# chronyc sources
MS Name/IP address         Stratum Poll Reach LastRx Last sample
===============================================================================
[root@client.user.net ~]#
```

3: (chronyc sources)

- chronyc sources
-
- ^* —

```
[root@client.user.net ~]# chronyc sources
MS Name/IP address         Stratum Poll Reach LastRx Last sample
===============================================================================
[root@client.user.net ~]#
```

3: (chronyc sources)

- chronyc sources
-
- ^* —
- ^+ —

```
[root@client.user.net ~]# chronyc sources
MS Name/IP address         Stratum Poll Reach LastRx Last sample
===============================================================================
[root@client.user.net ~]#
```

3: (chronyc sources)

- chronyc sources
-
- ^* —
- ^+ —
- ^? —

```
[root@client.user.net ~]# chronyc sources
MS Name/IP address         Stratum Poll Reach LastRx Last sample
===============================================================================
[root@client.user.net ~]#
```

3: (chronyc sources)

- chrony.conf allow 192.168.0.0/16

```
[user@server.user.net ~]$ sudo -i
[sudo] password for user:
[root@server.user.net ~]# hwclock
2025-11-22 16:55:49.068820+00:00
[root@server.user.net ~]# dnf -y install chrony
Extra Packages for Enterprise Linux 9 - x86_64   25 kB/s |  34 kB     00:01
Extra Packages for Enterprise Linux 9 - x86_64  788 kB/s |  20 MB     00:25
Rocky Linux 9 - BaseOS                          6.7 kB/s | 4.1 kB     00:00
Rocky Linux 9 - AppStream                       9.2 kB/s | 4.5 kB     00:00
Rocky Linux 9 - Extras                          7.8 kB/s | 2.9 kB     00:00
Package chrony-4.5-1.el9.x86_64 is already installed.
Dependencies resolved.
================================================================================
 Package          Architecture     Version               Repository       Size
================================================================================
Upgrading:
 chrony           x86_64           4.6.1-1.el9           baseos           340 k

Transaction Summary
================================================================================
Upgrade  1 Package

Total download size: 340 k
Downloading Packages:
chrony-4.6.1-1.el9.x86_64.rpm                   652 kB/s | 340 kB     00:00
--------------------------------------------------------------------------------
Total                                           387 kB/s | 340 kB     00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                        1/1
  Running scriptlet: chrony-4.6.1-1.el9.x86_64                              1/2
  Upgrading        : chrony-4.6.1-1.el9.x86_64                              1/2
  Running scriptlet: chrony-4.6.1-1.el9.x86_64                              1/2
  Running scriptlet: chrony-4.5-1.el9.x86_64                                2/2
  Cleanup          : chrony-4.5-1.el9.x86_64                                2/2
  Running scriptlet: chrony-4.5-1.el9.x86_64                                2/2
  Verifying        : chrony-4.6.1-1.el9.x86_64                              1/2
```

- chrony.conf allow 192.168.0.0/16
-

```
[user@server.user.net ~]$ sudo -i
[sudo] password for user:
[root@server.user.net ~]# hwclock
2025-11-22 16:55:49.068820+00:00
[root@server.user.net ~]# dnf -y install chrony
Extra Packages for Enterprise Linux 9 - x86_64   25 kB/s |  34 kB     00:01
Extra Packages for Enterprise Linux 9 - x86_64  788 kB/s |  20 MB     00:25
Rocky Linux 9 - BaseOS                          6.7 kB/s | 4.1 kB     00:00
Rocky Linux 9 - AppStream                       9.2 kB/s | 4.5 kB     00:00
Rocky Linux 9 - Extras                          7.8 kB/s | 2.9 kB     00:00
Package chrony-4.5-1.el9.x86_64 is already installed.
Dependencies resolved.
================================================================================
 Package          Architecture     Version              Repository       Size
================================================================================
Upgrading:
 chrony           x86_64           4.6.1-1.el9          baseos           340 k

Transaction Summary
================================================================================
Upgrade  1 Package

Total download size: 340 k
Downloading Packages:
chrony-4.6.1-1.el9.x86_64.rpm                   652 kB/s | 340 kB     00:00
--------------------------------------------------------------------------------
Total                                           387 kB/s | 340 kB     00:00
Running transaction check
Transaction check succeeded.
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Transaction test succeeded.
Running transaction
  Preparing        :                                                        1/1
  Running scriptlet: chrony-4.6.1-1.el9.x86_64                              1/2
  Upgrading        : chrony-4.6.1-1.el9.x86_64                              1/2
  Running scriptlet: chrony-4.6.1-1.el9.x86_64                              1/2
  Running scriptlet: chrony-4.5-1.el9.x86_64                                2/2
  Cleanup          : chrony-4.5-1.el9.x86_64                                2/2
  Running scriptlet: chrony-4.5-1.el9.x86_64                                2/2
  Verifying        : chrony-4.6.1-1.el9.x86_64                              1/2
```

- `systemctl restart chronyd`

```
[root@server.user.net ~]# chronyc sources
MS Name/IP address         Stratum Poll Reach LastRx Last sample
===============================================================================
^* ntp21.vniiftri.ru             2   6   377    33  +1358us[+3550us] +/- 8299us
^- 46.52.232.73                  2   6   377    54    -18ms[  -16ms] +/-  132ms
^- 87.103.245.205                2   6   377    58    -18ms[  -16ms] +/-   63ms
^- host46-148-143-150.broad>     2   6   377     5  -2134us[-2134us] +/-   26ms
[root@server.user.net ~]# nano / etc /chrone
[root@server.user.net ~]# nano / etc /chrone.conf
[root@server.user.net ~]# nano /etc/chrony.conf
[root@server.user.net ~]#
```

5: chronyd firewall NTP

- `systemctl restart chronyd`
- firewall `ntp`

```
[root@server.user.net ~]# chronyc sources
MS Name/IP address         Stratum Poll Reach LastRx Last sample
===============================================================================
^* ntp21.vniiftri.ru             2   6   377    33  +1358us[+3550us] +/- 8299us
^- 46.52.232.73                  2   6   377    54    -18ms[  -16ms] +/-  132ms
^- 87.103.245.205                2   6   377    58    -18ms[  -16ms] +/-   63ms
^- host46-148-143-150.broad>     2   6   377     5  -2134us[-2134us] +/-   26ms
[root@server.user.net ~]# nano / etc /chrone
[root@server.user.net ~]# nano / etc /chrone.conf
[root@server.user.net ~]# nano /etc/chrony.conf
[root@server.user.net ~]#
```

5: chronyd firewall NTP

- `systemctl restart chronyd`
- firewall `ntp`
-

```
[root@server.user.net ~]# chronyc sources
MS Name/IP address         Stratum Poll Reach LastRx Last sample
===============================================================================
^* ntp21.vniiftri.ru             2   6   377    33  +1358us[+3550us] +/- 8299us
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[root@server.user.net ~]# nano / etc /chrone
[root@server.user.net ~]# nano / etc /chrone.conf
[root@server.user.net ~]# nano /etc/chrony.conf
[root@server.user.net ~]#
```

5: chronyd firewall NTP

- chronyc sources

```
[root@server.user.net ~]# firewall-cmd --add-service=ntp --permanent
success
[root@server.user.net ~]#
```

6:

- chronyc sources
- NTP-

```
[root@server.user.net ~]# firewall-cmd --add-service=ntp --permanent
success
[root@server.user.net ~]#
```

6:

- chronyc sources
- NTP-
-

```
[root@server.user.net ~]# firewall-cmd --add-service=ntp --permanent
success
[root@server.user.net ~]#
```

6:

- chrony.conf server server.bahis.net iburst

```
success
[root@server.user.net ~]# firewall-cmd --reload
success
[root@server.user.net ~]#
```

7:

- chrony.conf server server.bahis.net iburst
- server

```
success
[root@server.user.net ~]# firewall-cmd --reload
success
[root@server.user.net ~]#
```

7:

- chrony.conf server server.bahis.net iburst
- server
- iburst

```
success
[root@server.user.net ~]# firewall-cmd --reload
success
[root@server.user.net ~]#
```

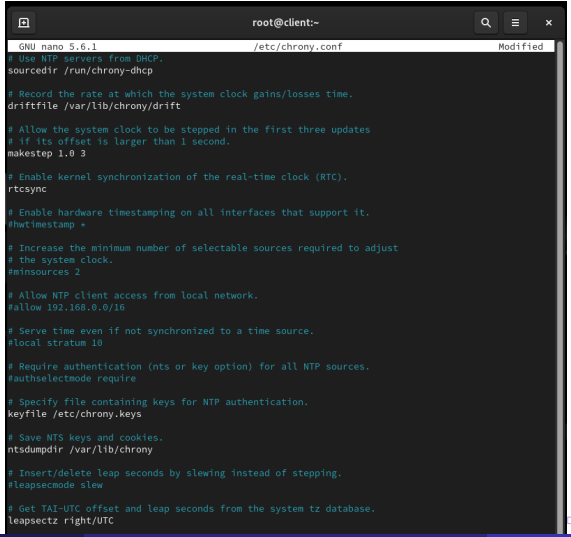
7:

- chronyd

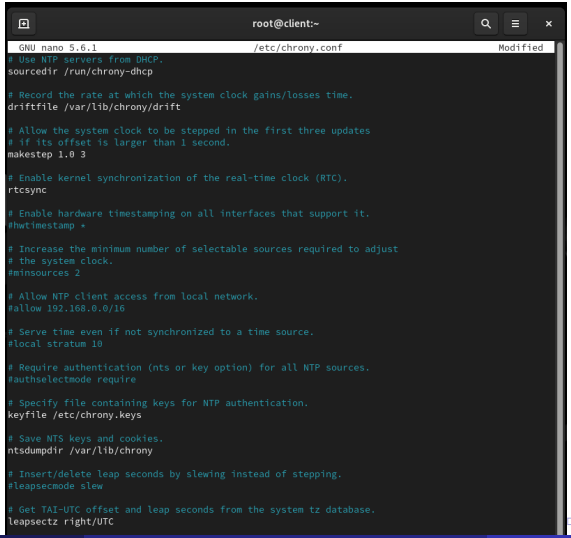
root@client:~

```
  GNU nano 5.6.1                    /etc/chrony.conf                    Modified
# Use NTP servers from DHCP.
sourcedir /run/chrony-dhcp

# Record the rate at which the system clock gains/losses time.
driftfile /var/lib/chrony/drift

# Allow the system clock to be stepped in the first three updates
# if its offset is larger than 1 second.
makestep 1.0 3

# Enable kernel synchronization of the real-time clock (RTC).
rtcsync

# Enable hardware timestamping on all interfaces that support it.
#hwtimestamp *

# Increase the minimum number of selectable sources required to adjust
# the system clock.
#minsources 2

# Allow NTP client access from local network.
#allow 192.168.0.0/16

# Serve time even if not synchronized to a time source.
#local stratum 10

# Require authentication (nts or key option) for all NTP sources.
#authselectmode require

# Specify file containing keys for NTP authentication.
keyfile /etc/chrony.keys

# Save NTS keys and cookies.
ntsdumpdir /var/lib/chrony

# Insert/delete leap seconds by slewing instead of stepping.
#leapsecmode slew

# Get TAI-UTC offset and leap seconds from the system tz database.
leapsectz right/UTC
```

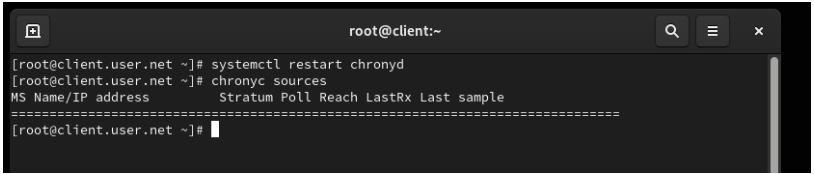
- `chronyd`
- (Stratum 4)



- `chronyd`
- (Stratum 4)
- `^*`

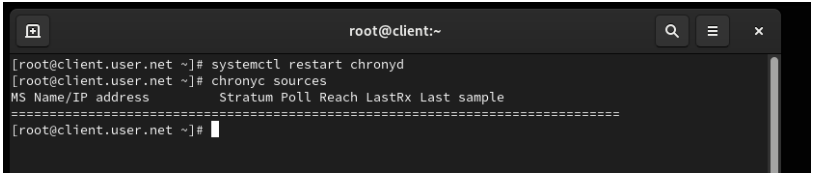


- /vagrant/provision/server ntp/etc



9: ntp chrony.conf

- /vagrant/provision/server ntp/etc
- chrony.conf



9: ntp chrony.conf

- ntp.sh

```
[root@server.user.net ~]# cd /vagrant/provision/server
[root@server.user.net server]# mkdir -p /vagrant/provision/server/ntp/etc
[root@server.user.net server]# cp -R /etc/chrony.conf /vagrant/provision/server/ntp/etc/
[root@server.user.net server]#
```

10: ntp.sh

- ntp.sh
-

```
[root@server.user.net ~]# cd /vagrant/provision/server
[root@server.user.net server]# mkdir -p /vagrant/provision/server/ntp/etc
[root@server.user.net server]# cp -R /etc/chrony.conf /vagrant/provision/server/ntp/etc/
[root@server.user.net server]#
```

10: ntp.sh

- ntp.sh chrony

```
[root@server.user.net server]# cd /vagrant/provision/server
[root@server.user.net server]# touch ntp.sh
[root@server.user.net server]# chmod +x ntp .sh
chmod: cannot access '.sh': No such file or directory
[root@server.user.net server]#
```

11: ntp.sh

- ntp.sh chrony
- /etc

```
[root@server.user.net server]# cd /vagrant/provision/server
[root@server.user.net server]# touch ntp.sh
[root@server.user.net server]# chmod +x ntp .sh
chmod: cannot access '.sh': No such file or directory
[root@server.user.net server]#
```

11: ntp.sh

- ntp.sh chrony
- /etc
- restorecon -vR /etc

```
[root@server.user.net server]# cd /vagrant/provision/server
[root@server.user.net server]# touch ntp.sh
[root@server.user.net server]# chmod +x ntp .sh
chmod: cannot access '.sh': No such file or directory
[root@server.user.net server]#
```

11: ntp.sh

- ntp.sh chrony
- /etc
- restorecon -vR /etc
- ntp firewall

```
[root@server.user.net server]# cd /vagrant/provision/server
[root@server.user.net server]# touch ntp.sh
[root@server.user.net server]# chmod +x ntp .sh
chmod: cannot access '.sh': No such file or directory
[root@server.user.net server]#
```

11: ntp.sh

- ntp.sh chrony
- /etc
- restorecon -vR /etc
- ntp firewall
- chronyd

```
[root@server.user.net server]# cd /vagrant/provision/server
[root@server.user.net server]# touch ntp.sh
[root@server.user.net server]# chmod +x ntp .sh
chmod: cannot access '.sh': No such file or directory
[root@server.user.net server]#
```

11: ntp.sh

- ntp/etc

root@server:/vagrant/provision/server

```
  GNU nano 5.6.1                                        ntp.sh
#!/bin/bash

echo "Provisioning script $0"

echo "Install needed packages"
dnf -y install chrony

echo "Copy configuration files"
cp -R /vagrant/provision/server/ntp/etc/* /etc

restorecon -vR /etc

echo "Configure firewall"
firewall-cmd --add-service=ntp
firewall-cmd --add-service=ntp --permanent

echo "Restart chronyd service"
systemctl restart chronyd
```

12: ntp chrony.conf

- ntp/etc
- chrony.conf

root@server:/vagrant/provision/server

```
  GNU nano 5.6.1                                        ntp.sh
#!/bin/bash

echo "Provisioning script $0"

echo "Install needed packages"
dnf -y install chrony

echo "Copy configuration files"
cp -R /vagrant/provision/server/ntp/etc/* /etc

restorecon -vR /etc

echo "Configure firewall"
firewall-cmd --add-service=ntp
firewall-cmd --add-service=ntp --permanent

echo "Restart chronyd service"
systemctl restart chronyd
```

12: ntp chrony.conf

- ntp.sh

```
[root@client.user.net ~]# cd /vagrant/provision/client
[root@client.user.net client]# touch ntp.sh
[root@client.user.net client]# chmod +x ntp.sh
[root@client.user.net client]#
```

13: ntp.sh

- ntp.sh
-

```
[root@client.user.net ~]# cd /vagrant/provision/client
[root@client.user.net client]# touch ntp.sh
[root@client.user.net client]# chmod +x ntp.sh
[root@client.user.net client]#
```

13: ntp.sh

- ntp.sh

root@client:/vagrant/provision/client

```
 GNU nano 5.6.1                                   ntp.sh
#!/bin/bash

echo "Provisioning script $0"

echo "Copy configuration files"
cp-R /vagrant/provision/client/ntp/etc/* /etc

restorecon-vR /etc

echo "Restart chronyd service"
systemctl restart chronyd
```

14: Vagrantfile

- `ntp.sh`
- `restorecon`

```
root@client:/vagrant/provision/client
GNU nano 5.6.1                                  ntp.sh
#!/bin/bash

echo "Provisioning script $0"

echo "Copy configuration files"
cp-R /vagrant/provision/client/ntp/etc/* /etc

restorecon-vR /etc

echo "Restart chronyd service"
systemctl restart chronyd
```

14: Vagrantfile

- ntp.sh
- restorecon
- chronyd

root@client:/vagrant/provision/client

```
 GNU nano 5.6.1                                   ntp.sh
#!/bin/bash

echo "Provisioning script $0"

echo "Copy configuration files"
cp-R /vagrant/provision/client/ntp/etc/* /etc

restorecon-vR /etc

echo "Restart chronyd service"
systemctl restart chronyd
```

14: Vagrantfile

- `Vagrantfile` provisioning-

- `Vagrantfile` provisioning-
- `provision/server/ntp.sh` `provision/client/ntp.sh`

- `Vagrantfile` provisioning-
- `provision/server/ntp.sh` `provision/client/ntp.sh`
- `preserve_order: true`

- `Vagrantfile` provisioning-
- `provision/server/ntp.sh` `provision/client/ntp.sh`
- `preserve_order: true`
- VM

3

3.

3.1

3.1

-
- `chronyd`

3.1

-
- `chronyd`
-

3.1

-
- `chronyd`
-
-

3.1

-
- `chronyd`
-
-
- `ntp` firewall

3.1

-
- `chronyd`
-
-
- `ntp` firewall
- provisioning Vagrant

3.1

-
- `chronyd`
-
-
- `ntp` firewall
- provisioning Vagrant
-